UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

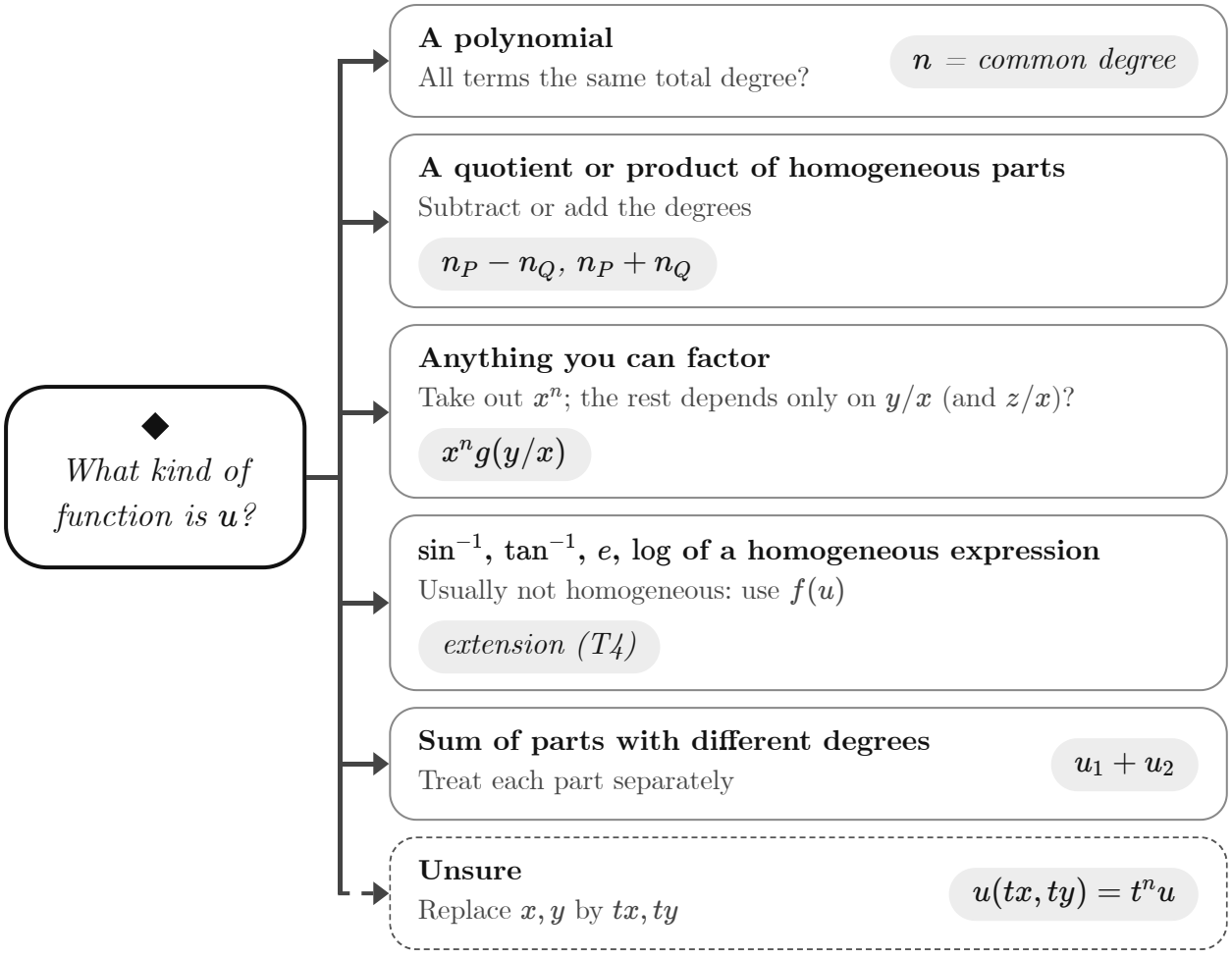
Class 005 · Homogeneous functions and Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

1. Figure F1.1 · Is it homogeneous, and of what degree?
2. Figure F3.1 · Evaluating Euler combinations for a sum of homogeneous parts.
3. Figure F4.1 · Using the extension of Euler's theorem.
4. Euler's theorem and its relatives

1 Homogeneous functions



Remark. The dashed route (the scaling test) is added for clarity; it always works and confirms the degree found by factoring.

Figure F1.1. Is it homogeneous, and of what degree? ESA ★★

S005·3 · S005·4 · T005 00:02:36 · T005 00:05:42

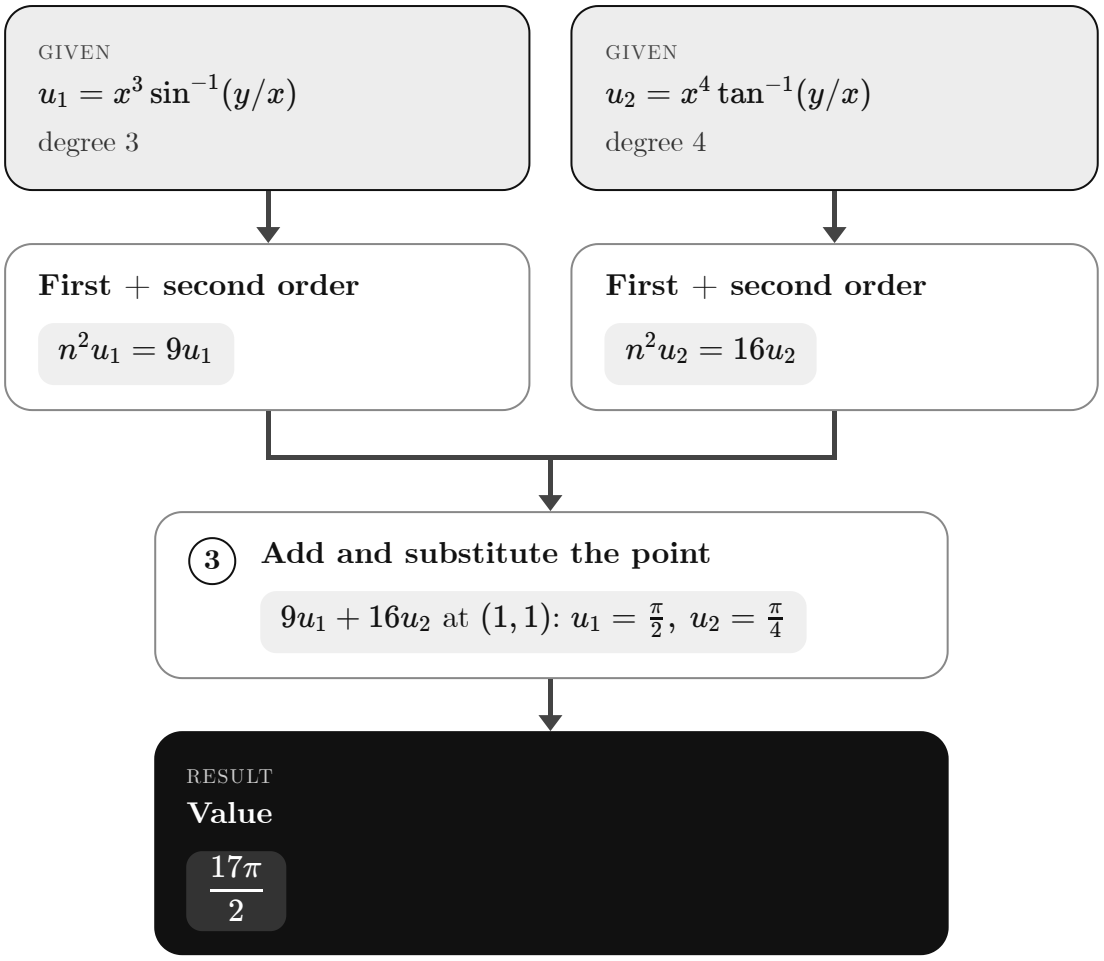
2 Euler's theorem

Euler's theorem and its relatives

	Applies when	Left side	Right side
Euler, first order	u homogeneous, degree n	$xu_x + yu_y$	nu
Corollary, second order	u homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$n(n-1)u$
Both added	u homogeneous, degree n	first + second order	n^2u
Extension, first order	$f(u)$ homogeneous, degree n	$xu_x + yu_y$	$g(u) = nf(u)/f'(u)$
Extension, second order	$f(u)$ homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$g(u)[g'(u) - 1]$
Three variables	$u(x, y, z)$ homogeneous	$xu_x + yu_y + zu_z$	nu

added for clarity S005·6 · S005·7

3 Second and higher order: the corollary



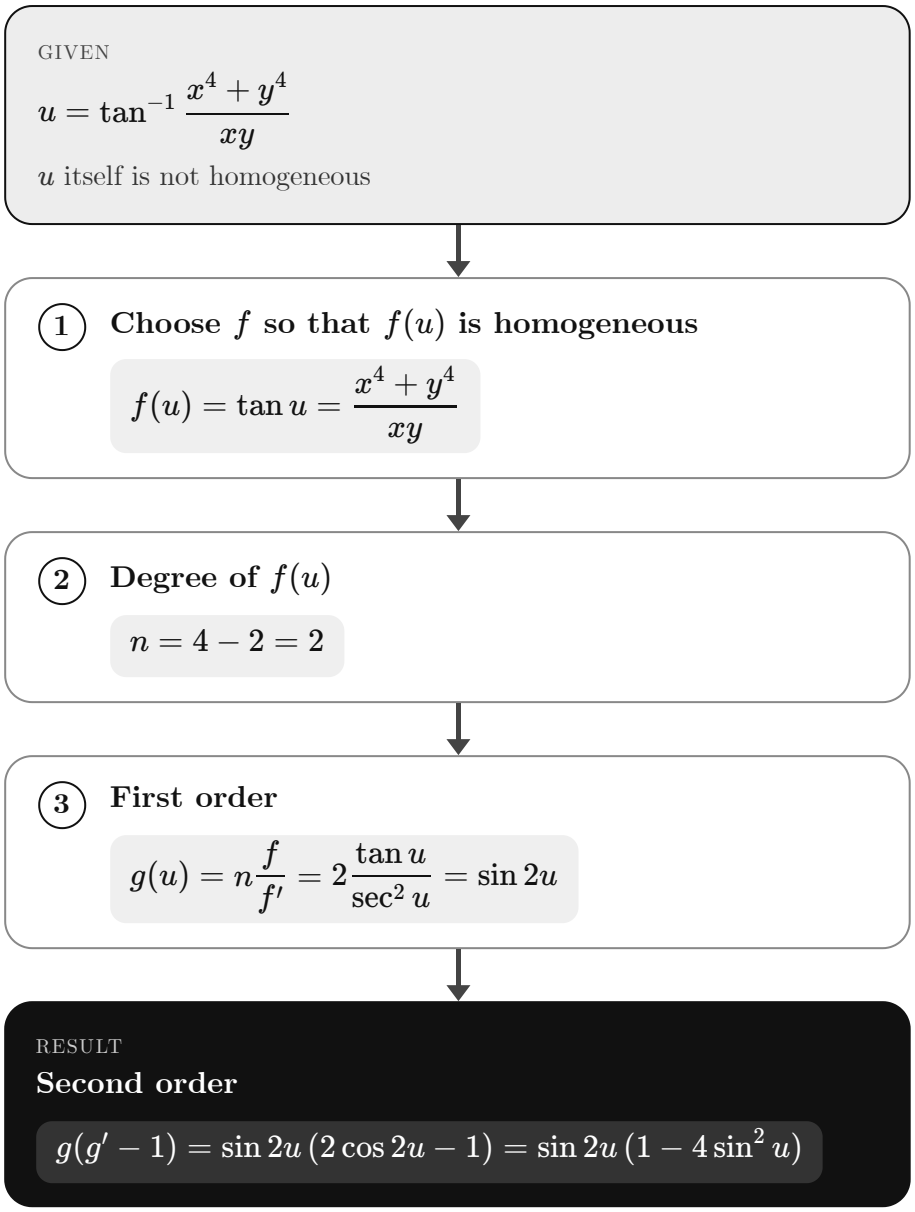
Remark. Only first order: n; only second order: $n(n-1)$; both: n^2. Read which combination the question asks for.

Figure F3.1. Evaluating Euler combinations for a sum of homogeneous parts.

ESA ★★★

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4 Extension: when f(u) is homogeneous



Remark. Common choices: $\sin^{-1}(\dots) \to f = \sin$ $(g = n\tan u)$; $\tan^{-1} \to f = \tan$ $(g = \frac{n}{2}\sin 2u)$; $\log(\dots) \to f = e^u$ $(g = n)$; $e^{(\cdots)} \to f = \log$ $(g = nu\log u)$.

Figure F4.1. Using the extension of Euler's theorem. ESA ★★★

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